

✗ Is 42 predefined?

No.

```
random_state=42
```

- 👉 42 is NOT predefined
- 👉 it is NOT a special ML value
- 👉 it is NOT required by Python or sklearn

It is just a **normal integer chosen by humans**

◆ Then what is random_state actually?

```
random_state
```

is a **parameter name** inside sklearn.

It controls:

```
how randomness behaves during splitting
```

◆ Why do we even need randomness here?

When you do:

```
train_test_split(X, y)
```

Python must:

```
shuffle data → then split 80/20
```

Because data is originally ordered like:

```
spam, spam, ham, ham, spam...
```

If you don't shuffle → bad training.

◆ What happens WITHOUT random_state?

Every time you run:

```
Run 1 → different split
Run 2 → different split
Run 3 → different split
```

So your model accuracy changes every time.

That is bad for:

- debugging
- comparison
- interviews
- reproducibility

♦ What happens WITH `random_state=42`?

```
Same input data
→ same shuffle pattern
→ same train/test split every time
```

So output becomes stable.

♦ Then why people use 42?

Just tradition.

It became popular because:

- easy to remember
- used in examples
- “The Hitchhiker’s Guide to the Galaxy” joke (42 = “answer to everything”)

But you could write:

```
random_state=1
random_state=0
random_state=100
random_state=999
```

All are valid.

They will still work the same way.

🔥 Simple analogy

Think:

```
Shuffling a deck of cards
```

- Without `random_state` → different shuffle every time
 - With `random_state=42` → same shuffle every time
-

✓ Final answer (interview-ready)

`random_state` is used to control randomness in data splitting. It ensures that the dataset is split in the same way every time the code runs. The value 42 is not special; it is just a commonly used fixed number for reproducibility.